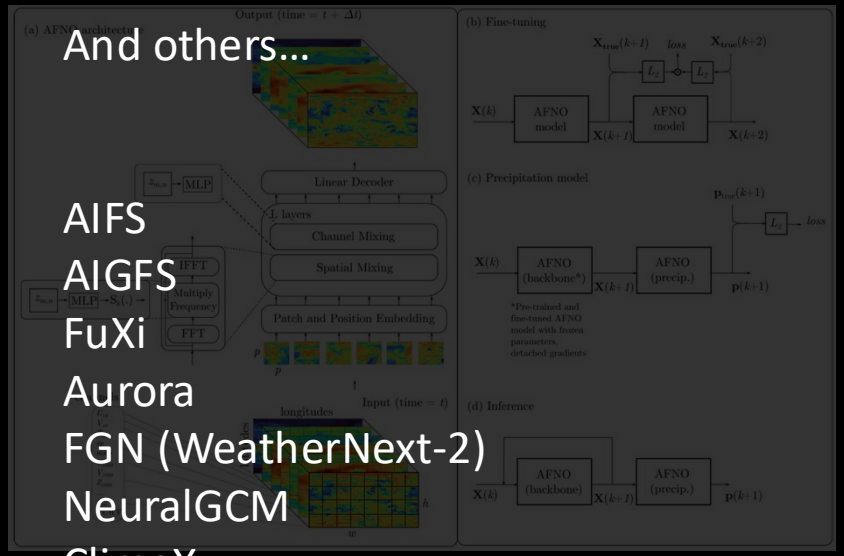
Model Evaluation: WeatherBench

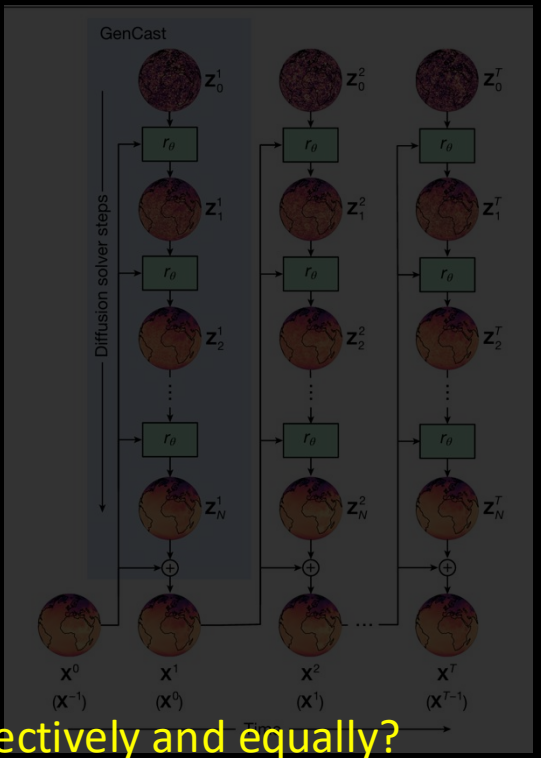
And others…

AIFS
AIGFS
FuXi
Aurora
FGN (WeatherNext-2)
NeuralGCM
ClimaX

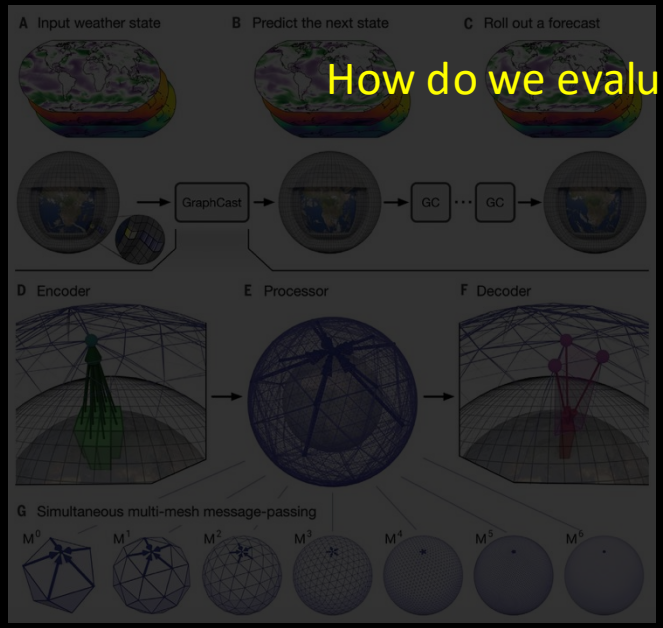


FourCastNet

GenCast

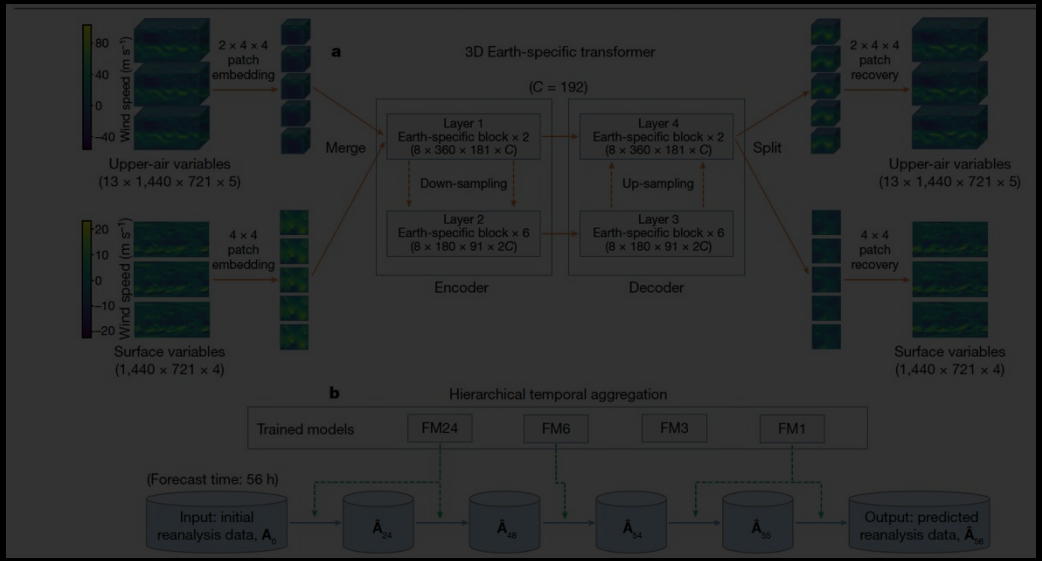


GraphCast



How do we evaluate all these models objectively and equally?

Pangu



WeatherBench*

Multiple metrics for deterministic and probabilistic model evaluations.

→ NOT a leaderboard like some benchmarks

→ Scores models on different variables, atmospheric levels, locations on globe

→ NOT continually updated. User can apply the open-source code to their own models or to existing model output for years not benchmarked.

*Rasp et al. paper describes WeatherBench-2. Code has been updated and is now branded as WeatherBench-X.

Deterministic Metrics

Root Mean Squared Error

$$\mathrm{RMSE}_l = \sqrt{\frac{1}{TIJ}\sum_t^T \sum_i^I \sum_j^J w(i)\left(f_{t,l,i,j} - o_{t,i,j}\right)^2}$$

Bias

$$\mathrm{Bias}_{l,i,j} = \frac{1}{T}\sum_t^T \left(f_{t,l,i,j} - o_{t,i,j}\right)$$

Anomaly Correlation Coefficient

$$\mathrm{ACC}_l = \frac{1}{T}\sum_t^T \frac{\sum_i^I \sum_j^J w(i) f'_{t,l,i,j} o'_{t,i,j}}{\sqrt{\sum_i^I \sum_j^J w(i) f'^{\,2}_{t,l,i,j} \sum_i^I \sum_j^J w(i) o'^{\,2}_{t,i,j}}}$$

$$f'_{t,l,i,j} = f_{t,l,i,j} - c_{t,l,i,j}; \quad o'_{t,i,j} = o_{t,i,j} - c_{t,i,j}$$

Ranges from -1 to 1. ACC < 0.6 is considered not usable forecast.

Stable Equitable Error in Probability Space (SEEPS): Used for precipitation

$$S = \frac{1}{2}\begin{bmatrix} 0 & \frac{1}{1-p_1} & \frac{4}{1-p_1} \\ \frac{1}{p_1} & 0 & \frac{3}{1-p_1} \\ \frac{1}{p_1} + \frac{3}{2+p_1} & \frac{3}{2+p_1} & 0 \end{bmatrix}$$

Probabilistic Metrics

Continuous Ranked Probability Score

$$CRPS_l := \frac{1}{T}\sum_{t}^{T}\left[\frac{1}{M}\sum_{m=1}^{M}\|f^{(m)} - o\|_{t,l} - \frac{1}{2M(M-1)}\sum_{m=1}^{M}\sum_{n=1}^{M}\|f^{(m)} - f^{(n)}\|_{t,l}\right].$$

General Form: $\mathrm{E}|X - Y| - 0.5E[X - X']$

E: Expected Value

X: Forecast

X': Another independent draw from forecast

Y: Observation

Spread/Skill (R = 1 means well-calibrated)

$$\mathrm{Spread}_l = \sqrt{\frac{1}{TIJ}\sum_{t}^{T}\sum_{i}^{I}\sum_{j}^{J} w(i)\mathrm{var}_m\left(f_{t,l,i,j,m}\right)}$$

$$R = \frac{\mathrm{Spread}}{\mathrm{RMSE}(\bar{f})}$$

Free Troposphere

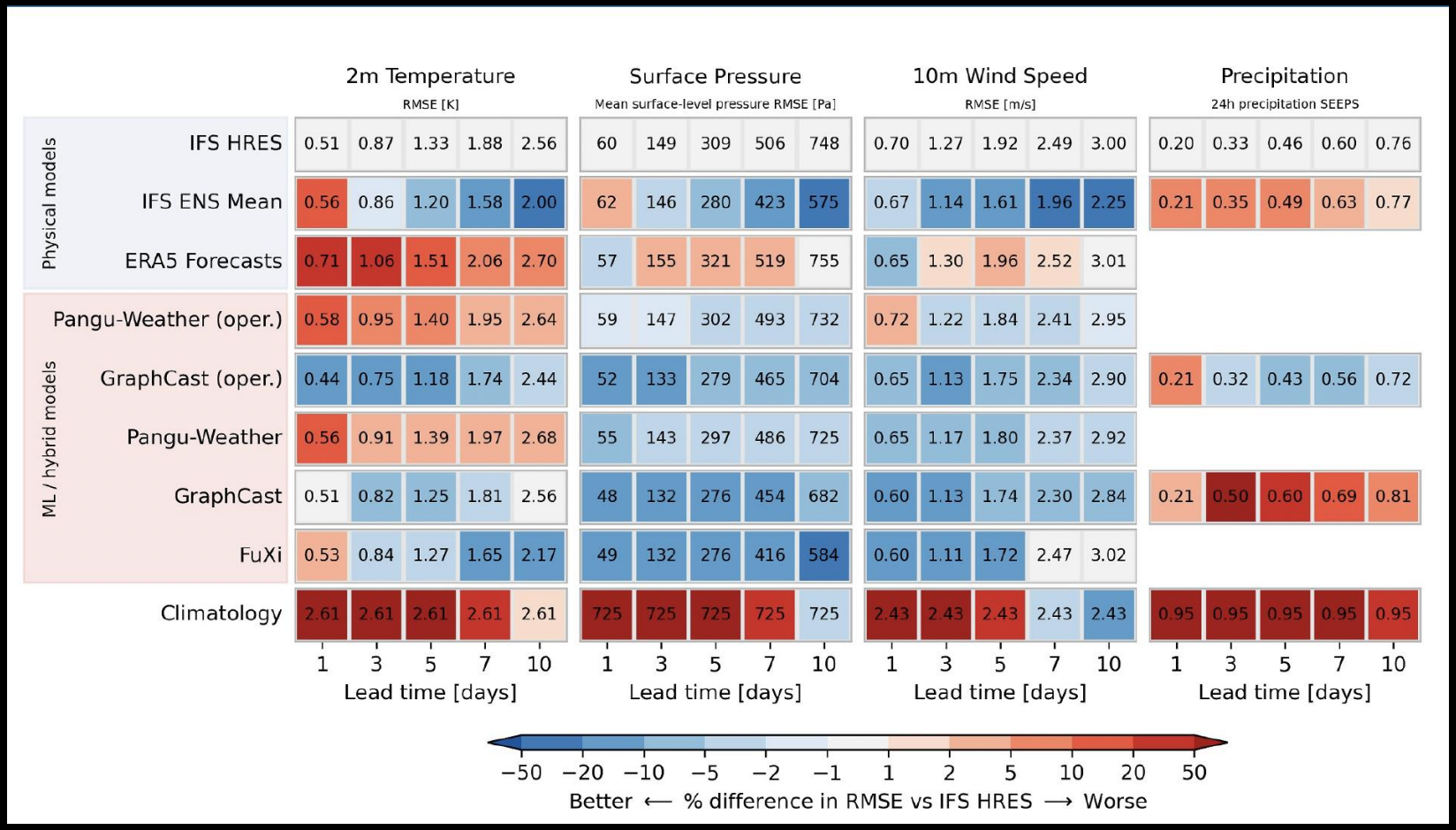
		Geopotential 500hPa geopotential RMSE [kg²/m²]					Temperature 850hPa temperature RMSE [K]					Humidity 700hPa specific humidity RMSE [g/kg]					Wind Vector 850hPa wind vector RMSE [m/s]				
		1	3	5	7	10	1	3	5	7	10	1	3	5	7	10	1	3	5	7	10
Physical models	IFS HRES	42	135	304	521	801	0.62	1.15	1.79	2.59	3.59	0.56	0.97	1.28	1.54	1.82	1.72	3.36	5.31	7.27	9.33
	IFS ENS Mean	42	132	277	439	621	0.63	1.09	1.59	2.14	2.77	0.52	0.85	1.07	1.23	1.39	1.66	3.04	4.53	5.85	7.08
	ERA5 Forecasts	43	142	316	534	811	0.59	1.17	1.84	2.64	3.62	0.54	1.02	1.34	1.60	1.87	1.66	3.47	5.48	7.41	9.43
ML / hybrid models	Pangu-Weather (oper.)	45	136	300	510	785	0.63	1.07	1.71	2.50	3.51	0.54	0.87	1.18	1.46	1.77	1.73	3.08	4.95	6.89	9.00
	GraphCast (oper.)	40	124	277	477	751	0.52	0.91	1.53	2.32	3.36	0.48	0.77	1.03	1.30	1.61	1.51	2.79	4.61	6.54	8.71
	Keisler (2022)	66	174	345	544	787	0.79	1.20	1.83	2.59	3.51	0.66	0.94	1.20	1.42	1.67	2.29	3.56	5.27	6.99	8.80
	Pangu-Weather	44	133	294	501	778	0.61	1.03	1.68	2.47	3.50	0.54	0.88	1.20	1.48	1.80	1.69	3.06	4.92	6.85	8.98
	GraphCast	39	124	274	468	731	0.51	0.93	1.53	2.29	3.30	0.47	0.80	1.06	1.31	1.60	1.45	2.81	4.53	6.34	8.34
	FuXi	40	125	276	433	631	0.54	0.95	1.56	2.11	2.87						1.50	2.85	4.59	5.76	7.17
	SphericalCNN	54	161	338	546	815	0.71	1.15	1.82	2.60	3.57	0.59	0.90	1.18	1.44	1.74	2.08	3.44	5.27	7.16	9.17
	NeuralGCM 0.7°	37	115	267	469	751	0.53	0.95	1.56	2.34	3.38	0.49	0.84	1.13	1.41	1.72	1.52	2.87	4.67	6.63	8.82
	NeuralGCM ENS Mean	43	126	266	424	606	0.63	0.99	1.50	2.06	2.72	0.54	0.81	1.02	1.20	1.38	1.78	2.93	4.36	5.70	6.97
	Climatology	820	820	820	820	820	3.41	3.41	3.41	3.41	3.41	1.61	1.61	1.61	1.61	1.61	8.05	8.05	8.05	8.05	8.05

Lead time [days]

−50 −20 −10 −5 −2 −1 1 2 5 10 20 50

Better ⟵ % difference in RMSE vs IFS HRES ⟶ Worse

Surface


2m Temperature
RMSE [K]
Surface Pressure
Mean surface-level pressure RMSE [Pa]
10m Wind Speed
RMSE [m/s]
Precipitation
24h precipitation SEEPS
Physical models
IFS HRES 0.51 0.87 1.33 1.88 2.56 60 149 309 506 748 0.70 1.27 1.92 2.49 3.00 0.20 0.33 0.46 0.60 0.76
IFS ENS Mean 0.56 0.86 1.20 1.58 2.00 62 146 280 423 575 0.67 1.14 1.61 1.96 2.25 0.21 0.35 0.49 0.63 0.77
ERA5 Forecasts 0.71 1.06 1.51 2.06 2.70 57 155 321 519 755 0.65 1.30 1.96 2.52 3.01
ML / hybrid models
Pangu-Weather (oper.) 0.58 0.95 1.40 1.95 2.64 59 147 302 493 732 0.72 1.22 1.84 2.41 2.95
GraphCast (oper.) 0.44 0.75 1.18 1.74 2.44 52 133 279 465 704 0.65 1.13 1.75 2.34 2.90 0.21 0.32 0.43 0.56 0.72
Pangu-Weather 0.56 0.91 1.39 1.97 2.68 55 143 297 486 725 0.65 1.17 1.80 2.37 2.92
GraphCast 0.51 0.82 1.25 1.81 2.56 48 132 276 454 682 0.60 1.13 1.74 2.30 2.84 0.21 0.50 0.60 0.69 0.81
FuXi 0.53 0.84 1.27 1.65 2.17 49 132 276 416 584 0.60 1.11 1.72 2.47 3.02
Climatology 2.61 2.61 2.61 2.61 2.61 725 725 725 725 725 2.43 2.43 2.43 2.43 2.43 0.95 0.95 0.95 0.95 0.95
1 3 5 7 10
Lead time [days]
−50 −20 −10 −5 −2 −1 1 2 5 10 20 50
Better ⟵ % difference in RMSE vs IFS HRES ⟶ Worse

Deterministic scorecards:
https://sites.research.google/gr/weatherbench/deterministic-scores/

Probabilistic scorecards:
https://sites.research.google/gr/weatherbench/probabilistic-scores/